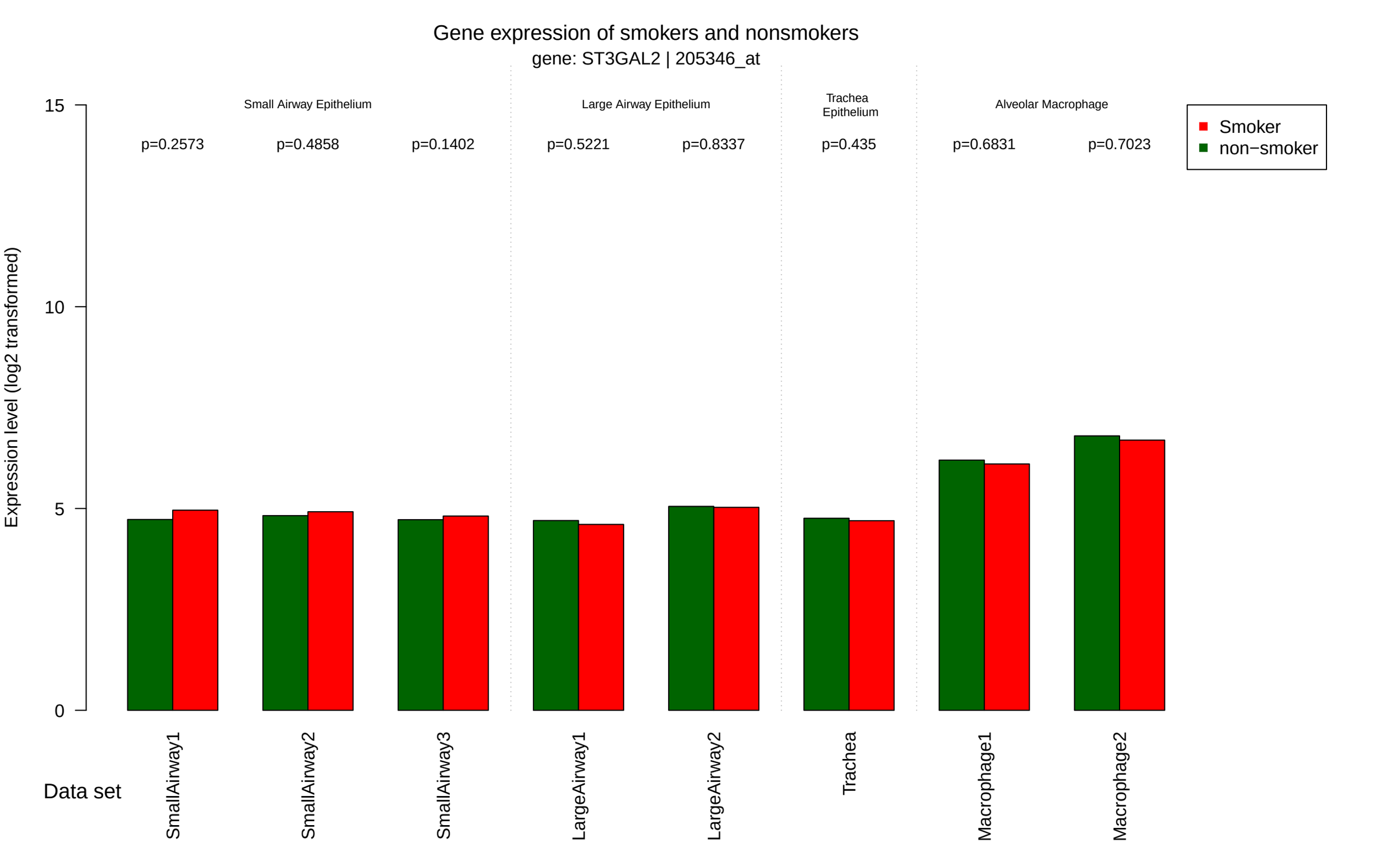
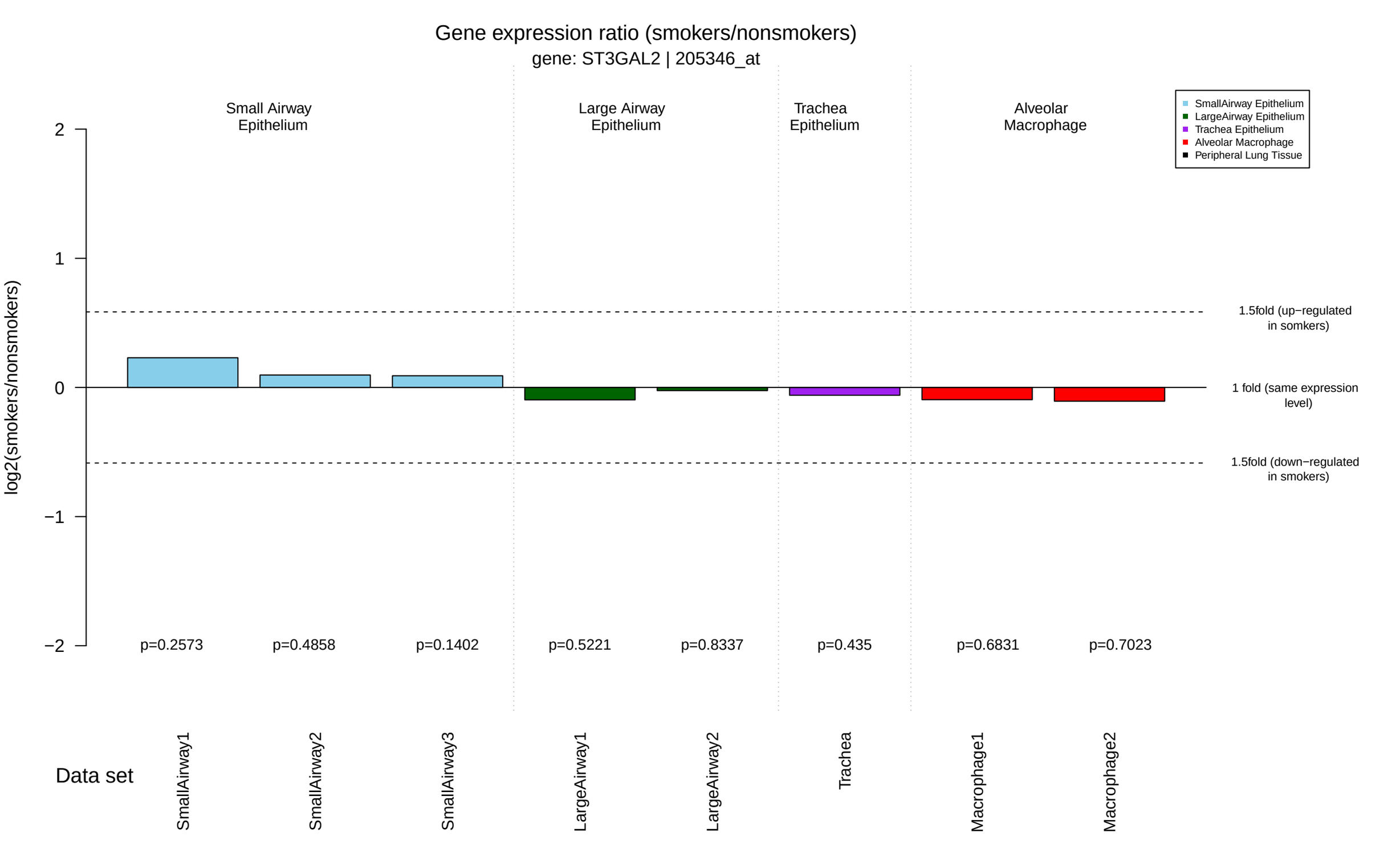
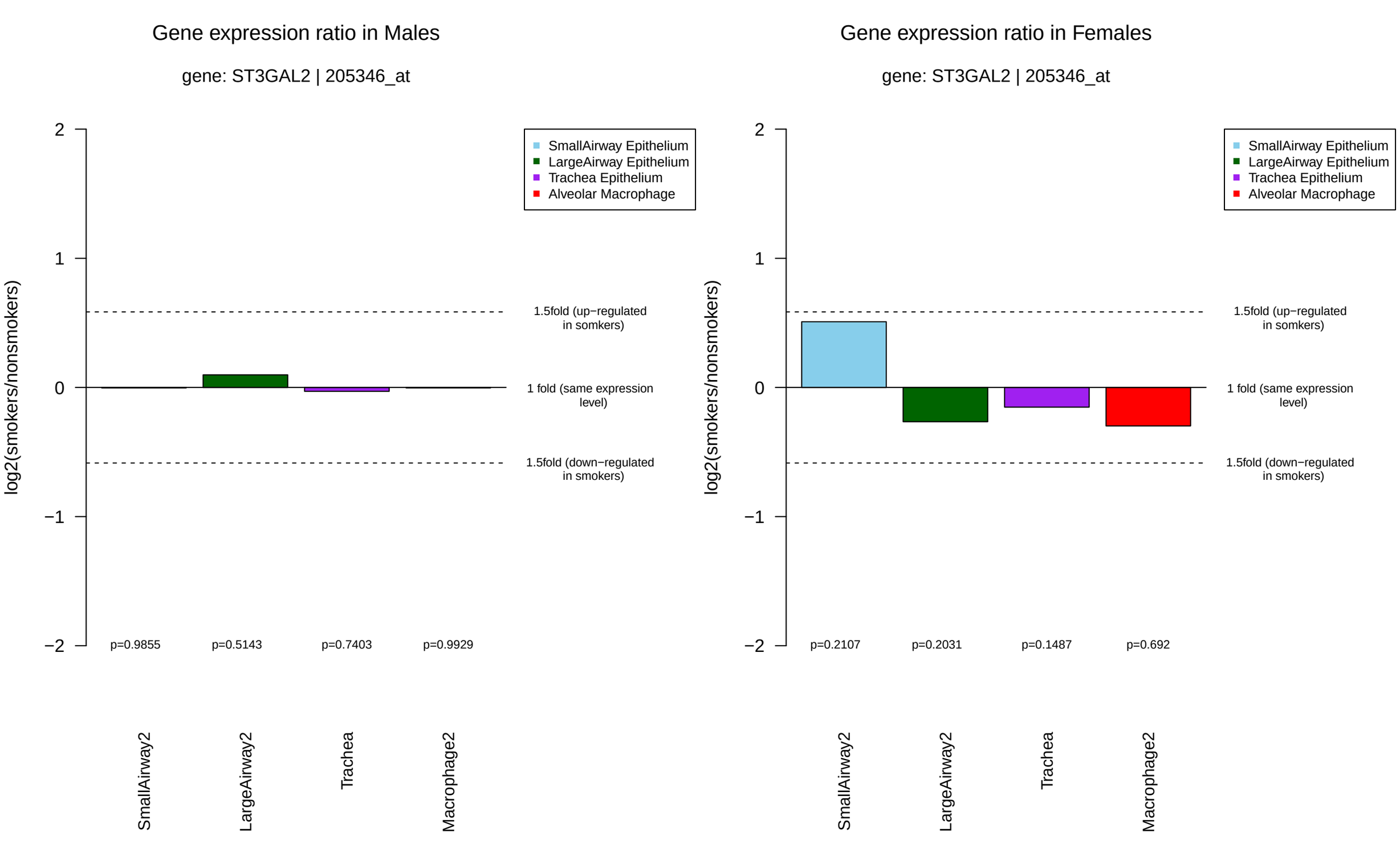
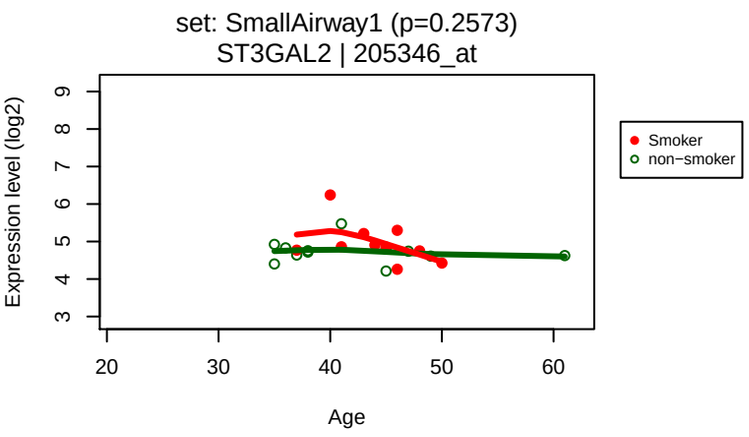
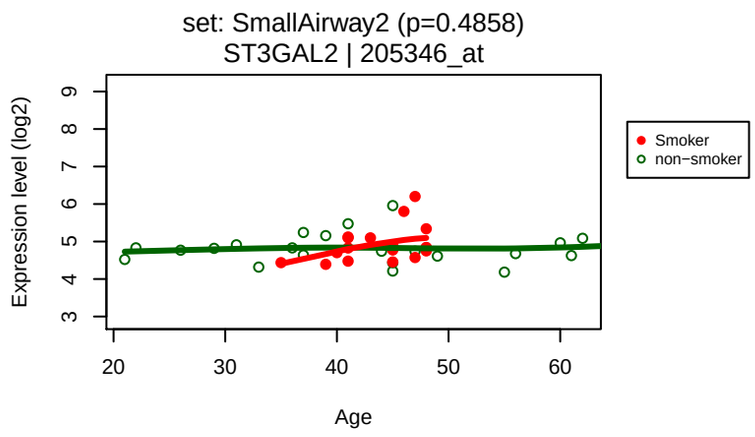
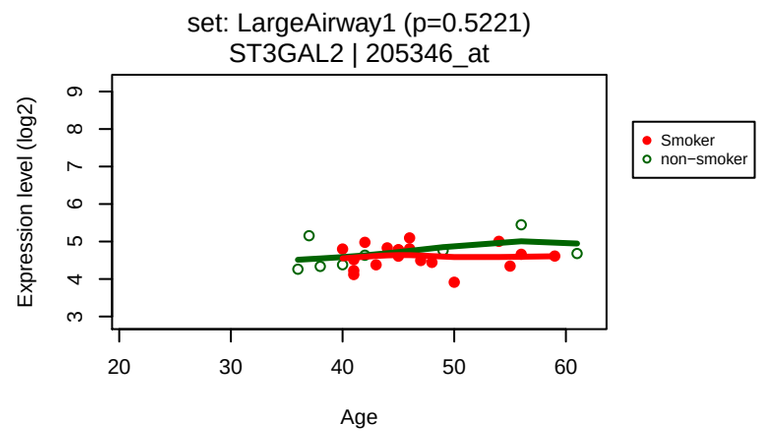

Gene expression of smokers and nonsmokers
gene: ST3GAL2 | 205346_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2573
p=0.4858
p=0.1402
p=0.5221
p=0.8337
p=0.435
p=0.6831
p=0.7023
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

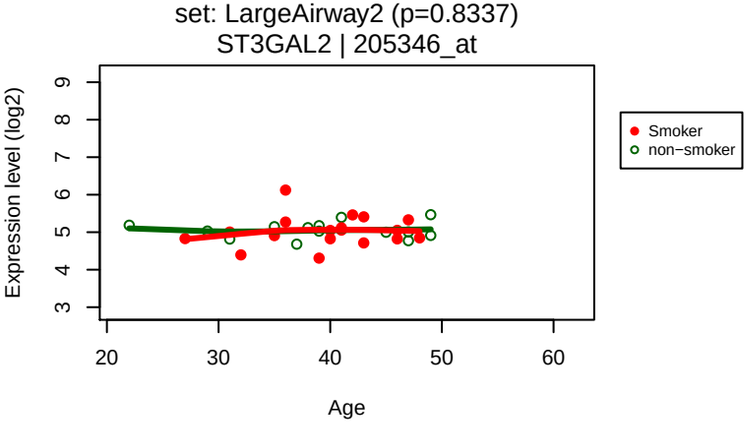

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL2 | 205346_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2573
p=0.4858
p=0.1402
p=0.5221
p=0.8337
p=0.435
p=0.6831
p=0.7023
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

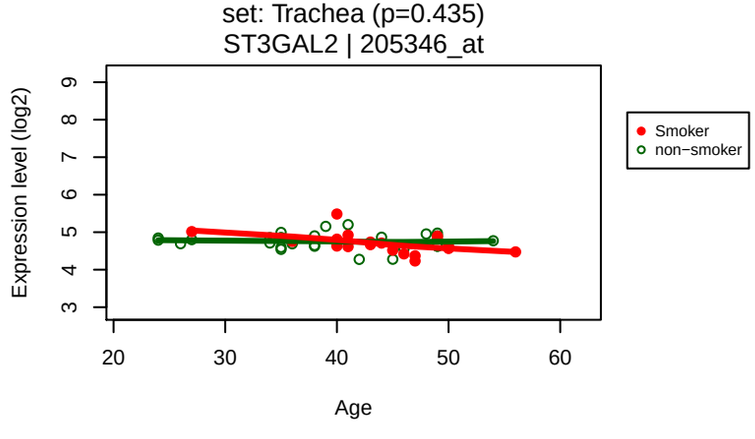

Gene expression ratio in Males
gene: ST3GAL2 | 205346_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9855
p=0.5143
p=0.7403
p=0.9929
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL2 | 205346_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2107
p=0.2031
p=0.1487
p=0.692
SmallAirway2
LargeAirway2
Trachea
Macrophage2

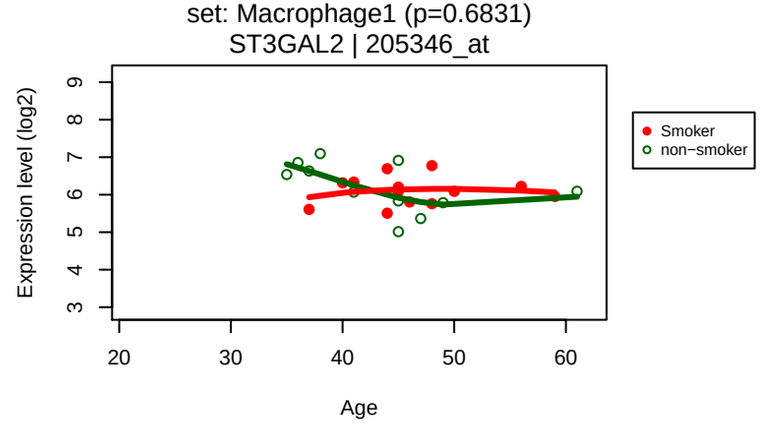

set: SmallAirway1 (p=0.2573)
ST3GAL2 | 205346_at
Expression level (log2)
Age
Smoker
non−smoker

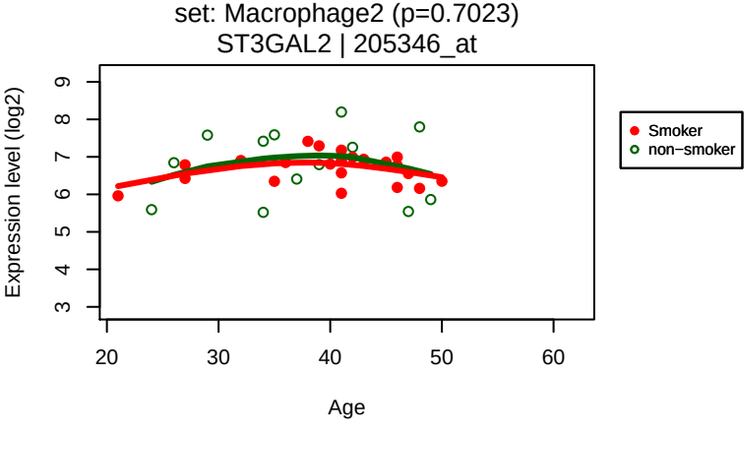

set: SmallAirway2 (p=0.4858)
ST3GAL2 | 205346_at
Expression level (log2)
Age
Smoker
non−smoker

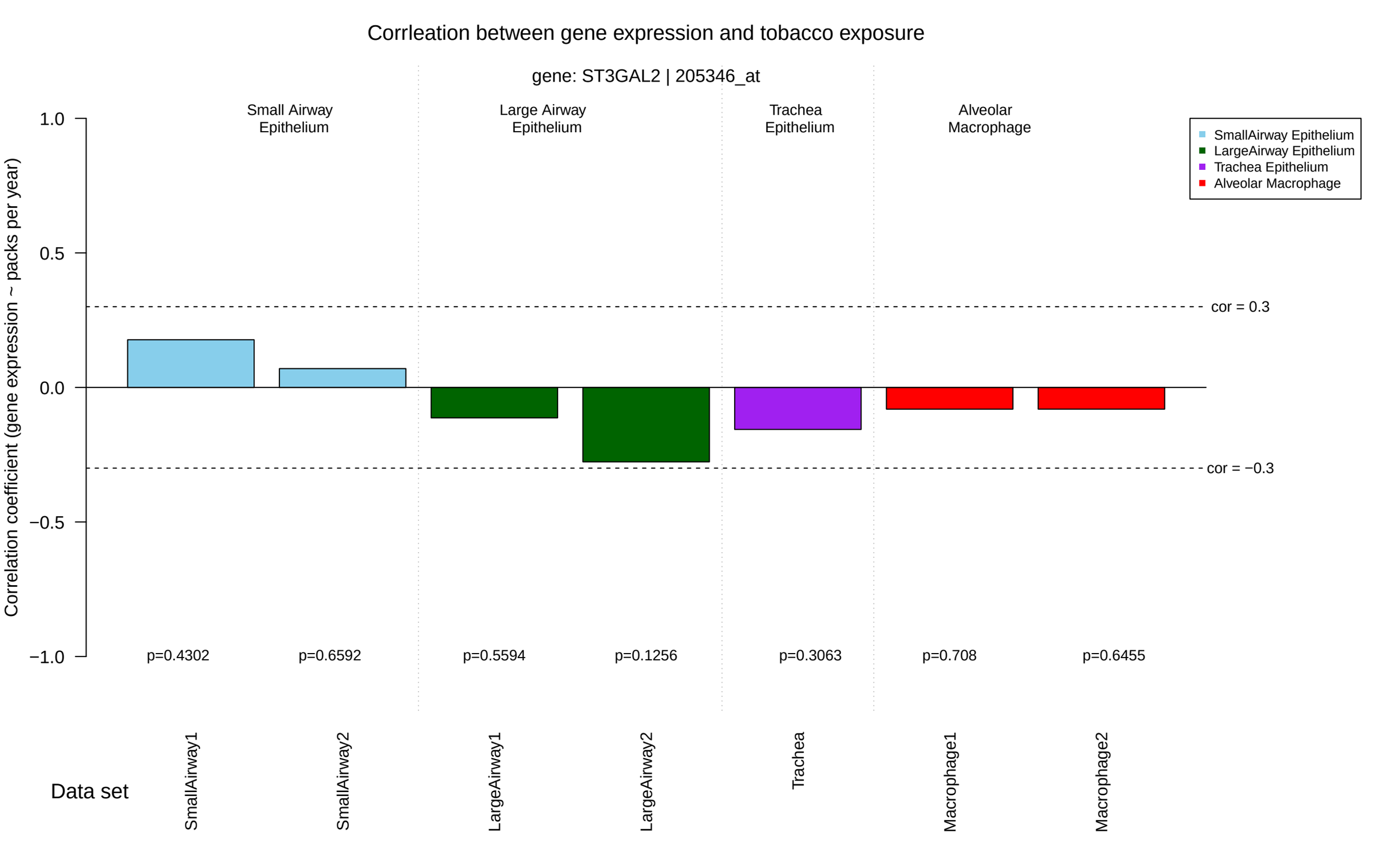

set: LargeAirway1 (p=0.5221)
ST3GAL2 | 205346_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8337)
ST3GAL2 | 205346_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.435)
ST3GAL2 | 205346_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6831)
ST3GAL2 | 205346_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7023)
ST3GAL2 | 205346_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL2 | 205346_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4302
p=0.6592
p=0.5594
p=0.1256
p=0.3063
p=0.708
p=0.6455
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2